The DH Approach to Forward Kinematics

Summary:

In this lesson, students find where the end of the arm is with vectors, using the DH convention that engineers use for robot arms. In the DH convention, each joint angle is measured from the link before it, and up (counterclockwise) is positive. On Day 1, students use the flat (2D) arm from Lesson 3. They write the upper arm as an arrow (cos θS, sin θS), turn that arrow 90° to get a second arrow at a right angle, and use the two arrows to find the direction of the forearm from the elbow angle. Adding the link vectors gives the tip. On Day 2, the base turn gives each arrow a third number, and students check their answers in the Desmos graph. At the end, students can find the end effector position for any joint angles.

Objective:

By the end of this lesson, students will be able to:

- Measure joint angles with the DH convention, and change Lesson 3 angles into DH angles
- Write a link as a unit vector (cos θ, sin θ) times the link's length
- Turn a vector 90° with (x, y) → (−y, x)
- Find the forearm's direction with f = cos θE · e + sin θE · n
- Add the link vectors in a link table to find the tip of the arm, in 2D and in 3D
- Check answers with the Desmos DH graph and explain what each part of the graph does

Standards:

NGSS

- Engineering Design: HS-ETS1-2 (break a complex problem into smaller problems), HS-ETS1-4 (use a computer model to test a solution)
- Science & Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking
- Crosscutting Concepts: Systems and System Models; Patterns

Common Core Math

- Vectors: HSN-VM.A.1 (a vector has a length and a direction), HSN-VM.B.4 (add vectors), HSN-VM.B.5 (multiply a vector by a number)
- Trigonometry: HSF-TF.A.2 (the unit circle), HSG-SRT.C.8 (use trig ratios to solve right triangles)

Vocabulary:

Forward kinematics: Finding where the end effector is from the joint angles.

End effector: The tip of the arm. It is the point we want to find.

Vector: An arrow with a length and a direction. We write it as numbers, like (3, 4) in 2D or (3, 4, 2) in 3D.

Vector sum: What you get when you add two vectors. Add the first numbers together, then the second numbers (and the third numbers in 3D).

Unit vector: A vector with a length of 1. At angle θ from the floor, it is (cos θ, sin θ).

Link vector: The vector for one link. It is the link's unit vector times the link's length.

DH convention: A standard way to measure the joints of a robot arm, named after Denavit and Hartenberg. Each joint angle is measured from the link before it, and counterclockwise (up) is positive.

Shoulder angle (θS): The angle of the upper arm, measured up from the floor.

Elbow angle (θE): The angle of the forearm, measured from the line of the upper arm. It is negative when the elbow bends down.

Base angle (θB): How far the base has turned from the x-axis, looking down from above.

Perpendicular: At a right angle (90°).

Turned arrow (n): The upper arm's unit vector turned 90° up. In 2D, (x, y) turns into (−y, x).

Frame: A set of arrows at right angles that moves with a link. The upper arm's frame is e, n and the elbow's axle.

Reach (R): How far the end effector is from the base, measured along the floor.

Link table: A table with one row for each link. It lists the length, the joint angle, the arrow and the link vector.

DH table: The table engineers use to describe an arm with the DH convention. The first three columns of our link table are a simple DH table.

Resources & Materials:

- Lesson 4 slideshow, Days 1 and 2 (33 slides; the slide numbers in the Procedures are from this slideshow)
- Computer/projector for showing the slideshow
- The Desmos graph, for Day 2 (on the projector or on student devices)
- Calculators set to degree mode
- Graph paper and rulers for the Day 1 vector drawings (optional)
- Printed copies of the blank link table on slide 15 (optional)
- The built 4-DOF arm from Lesson 2, to hold at θS = 45° and θE = −45° and show the direction of each arrow (optional)

Procedures:

Day 1: The Flat Arm with Vectors

- **Opening (3 min)** — Show the title slide and "Where Lesson 3 Left Off" (slides 1–2). In Lesson 3, triangles gave (X, Y) = (426.78, 176.78) for θ1 = 45° and θ2 = 45°. Today we get the same answer with vectors, and we measure the angles the way engineers do (the DH convention).

- **Vectors (6 min)** — Show “What Is a Vector?” and “Adding Vectors” (slides 3–4). A vector is an arrow with a length and a direction. (3, 4) means 3 across and 4 up, and its length is 5 (Pythagorean theorem). Two walks in a row add up: (3, 4) + (2, −1) = (5, 3). Multiplying by a number makes the arrow longer but keeps its direction: 2 × (3, 4) = (6, 8). Students draw the walks on graph paper.
- **Measuring angles the DH way (7 min)** — Show “Measuring Angles the DH Way” (slide 5). In the DH convention, each joint angle is measured from the link before it, and counterclockwise (up) is positive. The shoulder angle θS is measured up from the floor. The elbow angle θE is measured from the line of the upper arm, so an elbow that bends down has a negative angle. Our arm is θS = 45° and θE = −45°. To change Lesson 3 angles into DH angles: θS = 90° − θ1, θE = −θ2 and θB = θ3. Engineers use this rule because each joint only needs to know about the link before it, so it works the same way for any number of joints.
- **An arrow for each link (5 min)** — Show “An Arrow for Each Link” (slide 6). A unit vector is an arrow with length 1. At angle θ from the floor, it is (cos θ, sin θ), the same as a point on the unit circle. Multiply it by the link length to get the link vector. The upper arm is 250 × (0.7071, 0.7071) = (176.78, 176.78). These are ΔX1 and ΔY1 from Lesson 3.
- **Try It Yourself #1 (4 min)** — Students find the link vector for L = 250 mm at θ = 30° and at θ = 90° (slide 7). The answers are (216.51, 125.00) and (0.00, 250.00) (slide 8).
- **Turning an arrow 90° (4 min)** — Show “Turning an Arrow 90°” (slide 9). To turn an arrow 90° counterclockwise, swap the two numbers and change the sign of the first one: (x, y) → (−y, x). So (3, 4) turns into (−4, 3). Students can check with a drawing that the two arrows are at a right angle. The upper arm’s arrow e = (0.7071, 0.7071) turns into n = (−0.7071, 0.7071). Together, e and n are the upper arm’s frame.
- **The forearm’s arrow (12 min)** — Show “Finding the Forearm’s Arrow” (slide 10). Start with the unit circle: an arrow at angle θ is cos θ along x, then sin θ along y. The forearm’s angle θE is measured from the upper arm, not from the floor, so at the elbow we tilt the axes. e (along the upper arm) takes the place of x, and n (e turned 90°) takes the place of y. The same rule on the tilted axes gives the forearm’s arrow: f = cos θE · e + sin θE · n, which means cos θE along e, then sin θE along n. Then show “The Forearm Turns from the Upper Arm” and “Worked Example: The Forearm’s Arrow” (slides 11–12). Check the easy cases first: θE = 0° gives f = e, θE = 90° gives f = n, and θE = −90° gives f = −n. For our arm, the elbow bends down, so θE = −45° (not 45°). cos(−45°) = 0.7071 and sin(−45°) = −0.7071, so f = (0.5000, 0.5000) + (0.5000, −0.5000) = (1.0000, 0.0000). The forearm is flat. Students can check this another way: the forearm’s angle from the floor is θS + θE = 0°. Point out that the forearm only needs the upper arm’s frame and its own angle. This is the main idea of the DH convention.
- **Two vectors, one tip (3 min)** — Show “Two Vectors, One Tip” (slide 13). The upper arm vector is a = 250 × e = (176.78, 176.78) and the forearm vector is b = 250 × f = (250.00, 0.00). Put the arrows end to end: a + b = (426.78, 176.78), the same answer as Lesson 3.
- **The link table (5 min)** — Show “The Link Table” (slide 14). Each link gets a row with its length, its angle, its arrow and its vector. The row for n helps find f. Add the vector column to get the tip. The first three columns are like the DH table engineers use. Tell students to keep all the digits in the calculator and round only at the end. With 4-decimal arrows, 250 × 0.8660 gives 216.50 instead of 216.51.

- **Try It Yourself #2 and #3 (9 min)** — Students fill in a link table for θS = 60°, θE = −30° (slide 15) and for θS = 30°, θE = 0° (slide 17). The answers are (341.51, 341.51) and (433.01, 250.00) (slides 16 and 18). These are the two Try It Yourself problems from Lesson 3 (θ1 = 30°, θ2 = 30° and θ1 = 60°, θ2 = 0°), so students can compare with their triangle answers. In #3, θE = 0°, so f = e and the arm is straight.
- **Recap (2 min)** — Show "Day 1 Recap" (slide 19). Each angle is measured from the link before it. A link's arrow is (cos θ, sin θ), turning 90° is (x, y) → (−y, x), and the forearm's arrow is f = cos θE · e + sin θE · n. Tell students that tomorrow the base turns, so each arrow gets a third number.

Day 2: Adding the Base Turn (3D)

- **Recap (3 min)** — Show the "Day 2" slide (slide 20). Ask: What is (0.6, 0.8) turned 90°? ((−0.8, 0.6).) For θS = 60° and θE = −30°, what is the forearm's angle from the floor? (30°.) On Day 1 the tip was (across, up) = (426.78, 176.78). Today the base turns, so "across" can point any way on the floor. Call it the reach, R = 426.78, and call up the height, z = 176.78. From now on z points up, as in Desmos (Lesson 3 called the up direction Y).
- **The base turns the whole arm (4 min)** — Show "The Base Turns the Whole Arm" (slide 21). The base turns the whole arm by θB. Point to the arm's shadow on the floor: that is the reach R. Split the shadow along x and y: x = R cos θB and y = R sin θB, as in Lesson 3 (Lesson 3 called these X and Z). The height z stays the same.
- **The upper arm's arrow in 3D (6 min)** — Show "The Upper Arm's Arrow in 3D" (slide 22). Build e in three steps. From the side, e is a Day 1 arrow at angle θS: cos θS along the floor and sin θS up. From above, the floor part points at angle θB, so it splits like R on slide 21: x = cos θS cos θB and y = cos θS sin θB. Turning the base does not change the height: z = sin θS. So e = (cos θS cos θB, cos θS sin θB, sin θS). In the Desmos graph, this arrow is (a, b, c).
- **The turned arrow in 3D (6 min)** — Show "The Turned Arrow in 3D" (slide 23). Build n the same way. From the side, turn e by 90° with the Day 1 rule: (cos θS, sin θS) turns into (−sin θS, cos θS), so n goes −sin θS along the floor (back toward the base) and cos θS up. From above, split the floor part with θB, the same as for e. So n = (−sin θS cos θB, −sin θS sin θB, cos θS). Check: this is e with θS + 90° in place of θS. Together with the elbow's axle, e and n make the upper arm's frame.
- **The forearm and the tip in 3D (3 min)** — Show "The Forearm and the Tip in 3D" (slide 24). The rules from Day 1 do not change: f = cos θE · e + sin θE · n and tip = L1 e + L2 f. The base angle is already inside e and n.
- **Worked example (9 min)** — Show "Worked Example: Set Up" and "Worked Example: Adding Up" (slides 25–26). With θB = 45°, θS = 45° and θE = −45°: e = (0.5000, 0.5000, 0.7071), n = (−0.5000, −0.5000, 0.7071) and f = (0.7071, 0.7071, 0.0000). The upper arm is (125.00, 125.00, 176.78), the forearm is (176.78, 176.78, 0.00), and the tip is (301.78, 301.78, 176.78).
- **Try It Yourself #4 (5 min)** — Students solve θB = 30°, θS = 30°, θE = 0° (slide 27). θE = 0°, so f = e and the arm is straight. The answer is (375.00, 216.51, 250.00) (slide 28).
- **Try It Yourself #5 (7 min)** — Students solve θB = 30°, θS = 60°, θE = −90° (slide 29). This is Practice Problem 2 from Lesson 3 (θ1 = 30°, θ2 = 90°, θ3 = 30°) in DH angles. θE = −90°, so f = −n. The answer is (295.75, 170.75, 91.51) (slide 30). Lesson 3 wrote it as (X, Y, Z) = (295.75, 91.51, 170.75), because up was Y. Ask why the forearm's z is negative (it points 30° below the line of the floor: 60° − 90° = −30°). If time is short, give this one as homework.

- **Check it in Desmos (6 min)** — Show "Check It in Desmos" (slide 31), then open the Desmos graph. The sliders u, v and w are θB, θS and θE in degrees. Type the angles in and read d. The graph uses 2.5 for 250 mm, so d is in hundreds of mm and students multiply it by 100. For the worked example, d = (3.0178, 3.0178, 1.7678), which is (301.78, 301.78, 176.78) mm. Then students check #4, d = (3.7500, 2.1651, 2.5000), and #5, d = (2.9575, 1.7075, 0.9151). If d looks wrong, check the sign of w first. In the Vectors folder, click the gray circles next to the hidden arrows to show the frames at the shoulder and the elbow.
- **What the graph does (4 min)** — Show "What the Graph Does" (slide 32). Match each part of the graph to a step from class: (a, b, c) is e, $(-ac, -bc, a^2 + b^2) \div \sqrt{(1 - c^2)}$ is n, cos(w)(a, b, c) + sin(w) n is f, d = 2.5 f + 2.5 e is the tip, and the hidden $(b, -a, 0) \div \sqrt{(1 - c^2)}$ is the elbow's axle. The graph's n looks different from ours, but it is the same arrow: $\sqrt{(1 - c^2)}$ is cos θS, and dividing by it makes the length 1. That is also why v must stay below 90°. Each link is one vector.
- **Wrap-up (7 min)** — Show "What We Learned" (slide 33). Students do the exit ticket from the Assessment section. Tell them that in Lesson 5 they will write these steps as a Python program.

Assessment:

Students do the Try It Yourself problems in the slideshow (#1–#5) and check each answer on the answer slide. They can check #4 and #5 in the Desmos DH graph.

Use these questions as an exit ticket or a short quiz. The answers are in red.

- Lesson 3 used θ1 = 30° and θ2 = 30°. What are θS and θE in the DH convention?

Answer: θS = 90° − 30° = 60° and θE = −30°

- What is the unit vector at 60° from the floor?

Answer: (0.5000, 0.8660)

- A link is 250 mm long and 30° from the floor. What is its vector?

Answer: (216.51, 125.00)

- Turn the vector (2, 5) by 90° counterclockwise.

Answer: (−5, 2)

- The elbow bends down 45° from the line of the upper arm. Is θE positive or negative?

Answer: Negative: θE = −45°. In the DH convention, up (counterclockwise) is positive.

- What is the forearm's arrow f when θE = 0°? When θE = 90°?

Answer: f = e when θE = 0° (no turn), and f = n when θE = 90° (a quarter turn up).

- Find the tip of the flat arm for θS = 90° and θE = −90°.

Answer: (250.00, 250.00). The upper arm points straight up to (0, 250), and the forearm points straight out, because 90° − 90° = 0° from the floor.

- In the Desmos graph, d = (3.7500, 2.1651, 2.5000). Where is the tip in mm?

Answer: (375.00, 216.51, 250.00). The graph uses 2.5 for 250 mm, so multiply by 100.

- Why do engineers measure each joint from the link before it?

Answer: Every joint then works the same way, so a longer arm only needs one more row in the table and the same steps.